C PROGRAMMING — ASSIGNMENT 1

TASK 1: UNDERSTAND THE PROBLEM AND IDENTIFY THE DATA

CO1 • Bloom's Level: K4 – Analyse • Marks: 4

Assignment title: Design, Implement and Validate a Reliable Billing Program for a Rural Electricity Co-operative

Scenario: Grama Jyothi Rural Electricity Co-operative bills households using previous and current meter readings. The program must calculate consumption, energy charges, fixed charges and the total bill reliably, including meter rollover.

1. Problem analysis

The program accepts the household number, consumer category, previous meter reading and current meter reading. It validates the readings, calculates units consumed, applies the energy tariff slabs, adds the fixed charge and prints an itemised electricity bill.

2. Inputs, outputs and purpose

Type	Data item	Purpose / justification
Input	Household number	Identifies the household whose bill is being calculated.
Input	Consumer category (D for Domestic)	Records the category associated with the consumer.
Input	Previous meter reading	Starting meter-register value for the billing period.
Input	Current meter reading	Ending meter-register value for the billing period.
Computed	Units consumed	Difference between readings, adjusted if the 16-bit meter rolls over.
Computed	Energy charge	Charge based on the consumption slab rules.
Constant / input	Fixed charge	A fixed amount added to the energy charge; screenshot program uses ■50.00.
Output	Total bill	Sum of energy charge and fixed charge, displayed in rupees and paise.

3. Values that should be constants

The meter maximum (65,535), rollover modulus (65,536), slab boundaries (100 and 200 units), energy rates (■0.00, ■2.25 and ■4.50 per unit), and fixed charge (■50.00 in the provided program) should be defined in one place as named constants. Centralising these values makes tariff revisions easier and reduces the chance of updating only part of the program.

4. Possible value ranges and data types

Data	Possible values	Design consideration
Meter readings	0 to 65,535	A 16-bit unsigned register is non-negative and wraps after 65,535.
Units consumed	Normally non-negative; may cross rollover	Use unsigned arithmetic and account for rollover.
Energy / fixed / total charge	Money is non-negative; may contain paise	Store money as integer paise for exact arithmetic; use a sufficiently wide integer type.
Household number	Positive identifier	Choose an integer type large enough for the numbering system.
Category	A character such as D	A char is appropriate for a single-letter category code.

5. Rules for a correct bill

• Meter readings must be within 0–65,535 for the stated 16-bit register.
• Consumption must not become negative when one rollover occurs.
• The energy charge must follow the tariff slabs: first 100 units free; units 101–200 at ■2.25 per unit; units above 200 at ■4.50 per unit.
• Total bill = energy charge + fixed charge.
• The bill must be non-negative and formatted to two decimal places.
• Invalid readings or invalid input should be rejected with a clear message.

6. Assumptions

This solution assumes a 16-bit meter register with a maximum reading of 65,535 and no more than one rollover between readings. The supplied screenshot's program uses a ■50 fixed charge and treats the consumer category as a displayed character. The screenshot shows the calculation for a domestic consumer with previous reading 1200 and current reading 1530.

7. Alternative concept: store meter readings as text

If readings were stored as strings such as `"01530"`, the program would first validate the characters and convert the digit string to an integer before comparing or subtracting the readings. A string preserves leading zeros and can be useful when the original textual format matters, but arithmetic still requires conversion.

Representation	Time per reading	Space per reading
Integer	Comparison / subtraction O(1) for fixed-width integer types	O(1) fixed storage per reading
Text with d digits	Validation and conversion O(d); string comparison can also inspect up to d characters	O(d) characters per reading

For a fixed-width meter register, integer storage is simpler and more efficient for calculations. Text storage uses space proportional to the number of digits d and requires validation/conversion, but may be preferable when preserving the exact input format is important.

8. Execution evidence supplied



Figure 1. Supplied C program and execution output for household 1001.

9. Result analysis

The screenshot uses household number 1001, category D, previous reading 1200 and current reading 1530. Consumption is 1530 – 1200 = 330 units. Under the displayed program's slab calculation, the energy charge is (100 × ■2.25) + (130 × ■4.50) = ■225.00 + ■585.00 = ■810.00. Adding the fixed charge of ■50.00 gives a total bill of ■860.00, matching the screenshot.

Conclusion

The billing problem needs validated meter readings, rollover-aware consumption, centrally defined tariff values and an itemised output. Integer meter values and integer paise are recommended for reliable calculations. The provided execution demonstrates a 330-unit bill totalling ■860.00.